SPEC 003D · MORTGAGE-QC-PROD

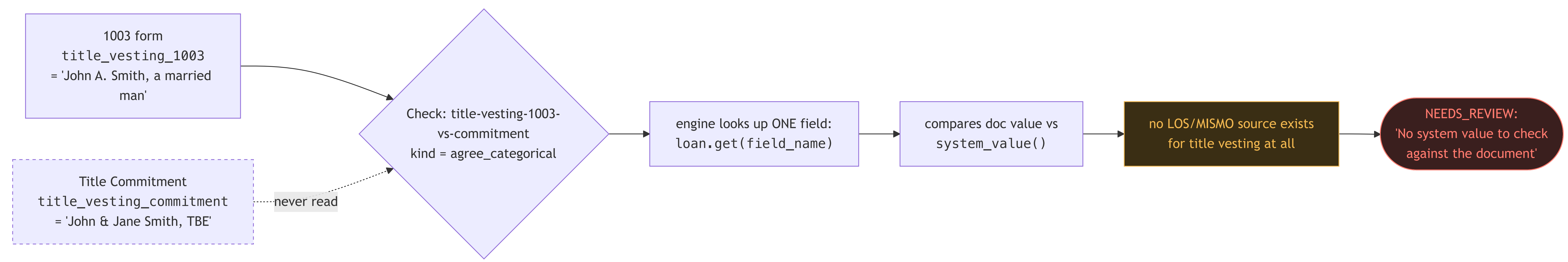
Doc-vs-doc reconciliation — how the fix actually flows

Four diagrams, in order: what's broken today, how the engine will evaluate the new check kinds, how the compiler decides which kind to use, and the two-phase rollout. Same plan as the text version — this is the shape of it.

01

The problem today

A loan has two documents that should agree on title vesting. The check exists and runs — but it only ever looks at *one* field, so the second document's value is never even read.

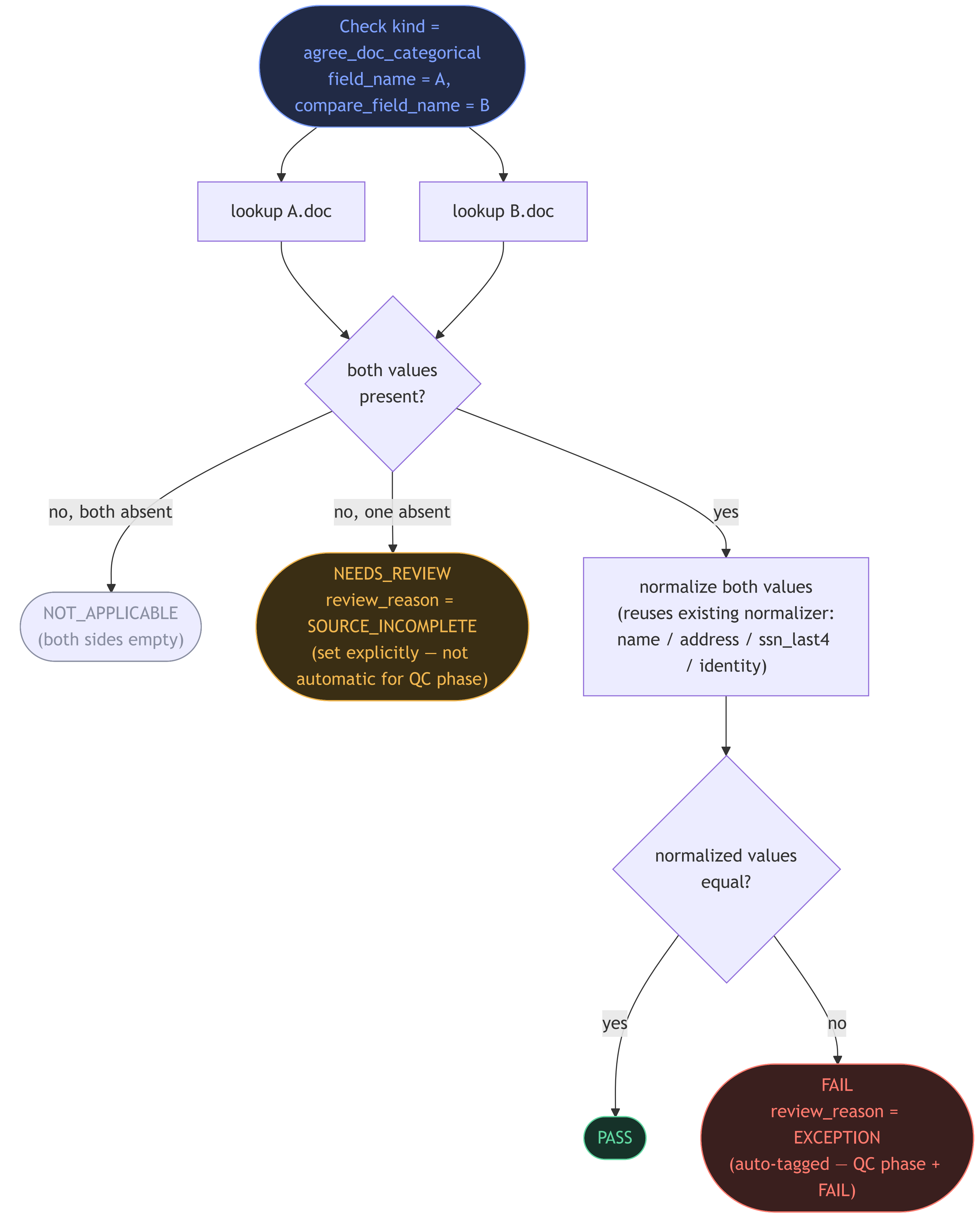


The real defect (the two documents disagree) is sitting right there — the engine's `agree_categorical` kind structurally can't see it, because it only ever compares one field's doc-value against that *same* field's system-value.

02

The fix — how the new check kind evaluates

Two new kinds, `agree_doc_categorical` and `agree_doc_numeric`, pull *two* fields and compare them directly — never touching the system/LOS/MISMO side at all.



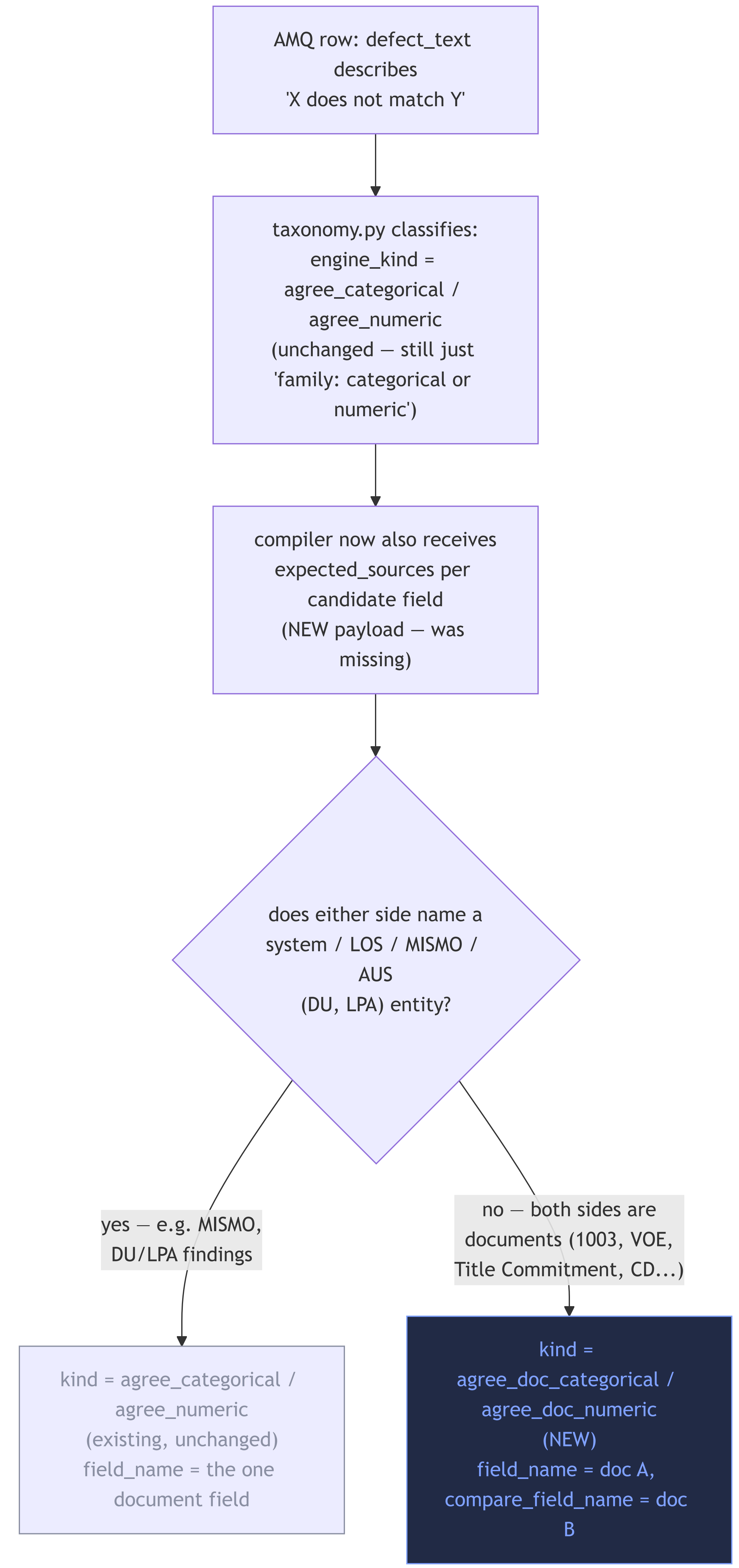
new logic · resolved cleanly · real defect surfaced · honest "can't tell yet"

`agree_doc_numeric` is the same shape, using tolerance instead of a normalizer — plus the same `UNSPECIFIED` -tolerance honesty guard the existing numeric kind already has.

03

How the compiler decides which kind to use

Nothing changes upstream in the row classifier — it still just says "this is a two-sided agreement check." The new decision (doc-vs-system vs. doc-vs-doc) happens at compile time, where the LLM has the full context to make the call.



AUS/DU/LPA output already counts as "system" by existing, signed precedent — that's not a new judgment call, just applying it consistently. The genuinely new signal is `expected_sources`: today the compiler doesn't even receive it, so it has no reliable way to tell "no system side exists" without this addition.

Rollout — two phases, kept deliberately separate

Phase 1 proves the mechanism for free against known ground truth. Phase 2 — spending real money to find more cases across the full rulebook — is a separate decision, not bundled in.

this plan's scope

Phase 1 — build & prove

Engine + schema + catalog + compiler changes. Validated by hand-authoring all 5 known doc-vs-doc checks directly against the existing loan fixtures — **zero LLM cost**. Closes 3 currently-broken checks into real `FAIL/PASS`, and 2 into a correct, explicit `NEEDS_REVIEW`.

separate decision, later

Phase 2 — recompile at scale

Full or partial recompile of the 8,442-row rulebook with the upgraded compiler, to catch the estimated 14–26 doc-vs-doc conditions across the real client rulebook, not just the 5 known synthetic ones. Real Bedrock spend — not committed here.

05

Files that change

Nine files, one new spec directory. No change to `model.py/SourceValue` at all — the whole point is that the new kinds never touch the system-source guard.

FILE	CHANGE
ruleset.py	Add `compare_field_name` field to `Check`
engine.py	Two new branches in `_eval_check`; explicit `SOURCE_INCOMPLETE` on one-side-absent
catalog.py	Referential integrity also resolves `compare_field_name`
compile_llm.py	Add `expected_sources` to payload; new source-shape decision rule; extend `_KIND_ONLY_FIELDS`
pattern_flags.py	Extend archetype-mismatch gate to the two new kinds
ruleset_defects.py	5 hand-authored checks for the 5 known defects; docstring correction
test_p0.py	Digest-bump handling (same pattern as feature 004)
test_doc_vs_doc_reconcile.py	New — engine-branch + phase/disposition tests
test_fixture_generation.py	Extend known-defects coverage from 20 to 25

full text plan: ~/.claude/plans/atomic-petting-floyd.md